General Assumptions (Assume for both the cases unless, me

Transactions in Year 1	141000
Average order size in Year 1	$258
COGS as a % of the sales price	70%
Annual growth rate of average order size	3%
Tax rate	35%
Discount rate (p.a.)	12%

Base case (No Web Portal) Assumptions

Annual growth rate of number of transactions	3%
Average processing cost per order	$30

Implementation of Web portal assumptions

Initial implementation cost	$4,000,000
Ongoing cost	$1,000,000
Total transaction increase in Year 1 from Base case	20000
Annual growth rate of number of transactions	10%
Average processing cost of a Web transaction	$3
Average processing cost per order	$16.50
% total transaction with the Web portal	50%

Conclu

entioned)

Base case (N

	Year 0	Year 1
Number of transactions	0	141000
Average order size (US $)		$ 258
Baseline Revenue (US $)	$ -	$ 36,378,000
COGS (US $)	$ -	$ 25,464,600
Order Processing Cost	$ -	$ 4,230,000
Net Income Before Tax		$ 6,683,400
Tax		$ 2,339,190
Net Income		$ 4,344,210
Cash Flows	$ -	$ 4,344,210

With the

		Year 0	Year 1
	Number of transactions	0	161000
	Average order size (US $)	$ -	$ 258
	Baseline Revenue (US $)	$ -	$ 41,538,000
	COGS (US $)	$ -	$29,076,600.00
	Order Processing Cost	$ -	$2,656,500.00
Costs of the Web portal	Initial implementation cost	$ 4,000,000	
	Ongoing cost	$ -	$1,000,000
	Net Income Before Tax	$ 4,000,000	$ 8,804,900
	Tax		$ 3,081,715
	Net Income	$ 4,000,000	$ 5,723,185
	Cash Flows	$ 4,000,000	$ 5,723,185

Incrementa

	Year 0	Year 1
Net incremental cash flows	$ 4,000,000	$ 1,378,975

Net present value (NPV)	$617,519.64

sion

Your conclusion: The datd shows that the web poral was efficent incressed revenue

No Web portal)

	Year 2		Year 3
	145230		$149,587
$	266	$	273.71
$	38,593,420.20		40943759.49
$	27,015,394	$	28,660,632
$	4,356,900	$	4,487,607
$	7,221,126	$	7,795,521
$	2,527,394	$	2,728,432
$	4,693,732	$	5,067,089
$	4,693,732	$	5,067,089

Web portal

	Year 2		Year 3
	177100		194810
$	266	$	274
$	47,062,554	$	53,321,874
$	32,943,788	$	37,325,312
$	2,922,150	$	3,214,365
$	1,000,000	$	1,000,000
$	10,196,616	$	11,782,197
$	3,568,816	$	4,123,768.99
$	6,627,801	$	7,658,428
$	6,627,801	$	7,658,428

al Cash Flows

	Year 2		Year 3
$	1,934,069	$	2,591,340

Retirement Savings

Assumptions			
Current Earnings	**$175,000**	Mr. Clark's Contribution To 401(k)	**6%**
Earnings Increase	**3%**	Employer Contribution to 401(k)	**5%**
Stock MKT return	**8%**	Additional savings	**$10,000**
		Increase in add. Savings	**4%**

		401(k) Plan			
Mr. Clark's Age	**Salary**	**Previous Year's Balance**	**Mr. Clark's Contribution**	**Employer Contribution**	**410(k) Stock Market Earnings**
40	**$175,000**	**$225,000**	**$10,500**	**$8,750**	**$19,540**
41	**$180,250**	**$263,790**	**$10,815**	**$9,013**	**$22,689**
42	**$185,658**	**$306,307**	**$11,139**	**$9,283**	**$26,138**
43	**$191,227**	**$352,868**	**$11,474**	**$9,561**	**$29,912**
44	**$196,964**	**$403,815**	**$11,818**	**$9,848**	**$34,038**
45	**$202,873**	**$459,519**	**$12,172**	**$10,144**	**$38,547**
46	**$208,959**	**$520,382**	**$12,538**	**$10,448**	**$43,469**
47	**$215,228**	**$586,837**	**$12,914**	**$10,761**	**$48,841**
48	**$221,685**	**$659,353**	**$13,301**	**$11,084**	**$54,699**
49	**$228,335**	**$738,437**	**$13,700**	**$11,417**	**$61,084**
50	**$235,185**	**$824,639**	**$14,111**	**$11,759**	**$68,041**
51	**$242,241**	**$918,550**	**$14,534**	**$12,112**	**$75,616**
52	**$249,508**	**$1,020,812**	**$14,970**	**$12,475**	**$83,861**
53	**$256,993**	**$1,132,119**	**$15,420**	**$12,850**	**$92,831**
54	**$264,703**	**$1,253,219**	**$15,882**	**$13,235**	**$102,587**
55	**$272,644**	**$1,384,923**	**$16,359**	**$13,632**	**$113,193**
56	**$280,824**	**$1,528,107**	**$16,849**	**$14,041**	**$124,720**
57	**$289,248**	**$1,683,718**	**$17,355**	**$14,462**	**$137,243**
58	**$297,926**	**$1,852,778**	**$17,876**	**$14,896**	**$150,844**
59	**$306,864**	**$2,036,393**	**$18,412**	**$15,343**	**$165,612**
60	**$316,069**	**$2,235,760**	**$18,964**	**$15,803**	**$181,642**
61	**$325,552**	**$2,452,170**	**$19,533**	**$16,278**	**$199,038**
62	**$335,318**	**$2,687,019**	**$20,119**	**$16,766**	**$217,912**
63	**$345,378**	**$2,941,817**	**$20,723**	**$17,269**	**$238,385**
64	**$355,739**	**$3,218,193**	**$21,344**	**$17,787**	**$260,586**
65	**$366,411**	**$3,517,910**	**$21,985**	**$18,321**	**$284,657**

Sensitivity Analysis

	Increase in earnings				
ket Return	$5,720,011	2%	3%	4%	5%
	4%	$2,680,458	$2,813,297	$2,967,929	$3,148,280
	6%	$3,819,174	$3,979,570	$4,164,968	$4,379,739

Average Mark

8%	$5,524,315	$5,720,011	$5,944,581	$6,202,910
10%	$8,083,007	$8,324,227	$8,599,020	$8,912,868
12%	$11,924,137	$12,224,418	$12,564,006	$12,949,082
15%	$21,565,125	$21,989,457	$22,464,234	$22,996,924

401K Plan Total	Personal Account Previous Year Balance	Additional Savings	P. Acc. Stock Market Earnings	Acct Total	Yearly total
$263,790	$85,000	$10,000	$7,600	$102,600	$366,390
$306,307	$102,600	$10,400	$9,040	$122,040	$428,347
$352,868	$122,040	$10,816	$10,628	$143,484	$496,352
$403,815	$143,484	$11,249	$12,379	$167,112	$570,927
$459,519	$167,112	$11,699	$14,305	$193,115	$652,634
$520,382	$193,115	$12,167	$16,423	$221,704	$742,086
$586,837	$221,704	$12,653	$18,749	$253,106	$839,943
$659,353	$253,106	$13,159	$21,301	$287,567	$946,920
$738,437	$287,567	$13,686	$24,100	$325,352	$1,063,790
$824,639	$325,352	$14,233	$27,167	$366,752	$1,191,391
$918,550	$366,752	$14,802	$30,524	$412,079	$1,330,629
$1,020,812	$412,079	$15,395	$34,198	$461,672	$1,482,484
$1,132,119	$461,672	$16,010	$38,215	$515,897	$1,648,015
$1,253,219	$515,897	$16,651	$42,604	$575,151	$1,828,370
$1,384,923	$575,151	$17,317	$47,397	$639,865	$2,024,788
$1,528,107	$639,865	$18,009	$52,630	$710,505	$2,238,612
$1,683,718	$710,505	$18,730	$58,339	$787,573	$2,471,291
$1,852,778	$787,573	$19,479	$64,564	$871,616	$2,724,394
$2,036,393	$871,616	$20,258	$71,350	$963,225	$2,999,618
$2,235,760	$963,225	$21,068	$78,743	$1,063,037	$3,298,797
$2,452,170	$1,063,037	$21,911	$86,796	$1,171,744	$3,623,914
$2,687,019	$1,171,744	$22,788	$95,563	$1,290,094	$3,977,113
$2,941,817	$1,290,094	$23,699	$105,103	$1,418,896	$4,360,713
$3,218,193	$1,418,896	$24,647	$115,483	$1,559,027	$4,777,220
$3,517,910	$1,559,027	$25,633	$126,773	$1,711,433	$5,229,343
$3,842,872	$1,711,433	$26,658	$139,047	$1,877,139	$5,720,011

6%	7%	8%
$3,358,998	$3,605,586	$3,894,561
$4,629,039	$4,918,960	$5,256,694

$6,500,744	$6,844,850	$7,243,193
$9,272,203	$9,684,571	$10,158,828
$13,386,874	$13,885,835	$14,455,851
$23,596,212	$24,272,195	$25,036,628

Student ID	Student Name	Major	Year at College	Eligible for CPA	Eligible for President
ABC123	Abby Cook	Finance	Freshmen	NO	NO
BRE234	Brian Edward	Accounting	Junior	NO	NO
SER456	Sam Johnson	MIS	Sophomore	NO	NO
RER4444	Ramesh Reddy	MKT	Freshmen	NO	NO
PER234	Peter Parker	Management	Senior	NO	YES
FED55555	Fred Dicker	Supply Chain	Senior	NO	YES
HGF6678	Harry Potter	Accounting	Sophomore	NO	NO
JOG68745	John Edward	MKT	Junior	NO	NO
KDFEGH4	Kaitlyn Xu	Management	Junior	NO	NO
SDEF345	Samantha Burrow	Finance	Senior	YES	NO
GHJ67	Grace Willams	MIS	Freshmen	NO	NO
LSVG345	Lisa Miller	Supply Chain	Senior	NO	YES

DDifference between Absolute and Relative References: Relative

Eligible to Mentor	GPA	Eligible for Honors
NO	3.5	NO
NO	2.3	NO
NO	3.7	NO
NO	3.9	NO
NO	3.1	NO
YES	3.9	YES
NO	2.9	NO
YES	3.8	NO
YES	3.75	NO
NO	2.3	NO
NO	3.9	NO
YES	3.85	YES

Reference refer

Difference betweer

n VLOOKUP and HLOOKUP Function : The dif